Coordinate methods and triangle congruence

Core Concepts: Coordinate methods and triangle congruence

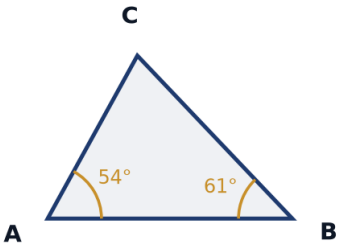
1 In $\triangle ABC$ studying coordinate methods and triangle congruence, $m\angle A = 34^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 101°

b. 111°

c. 93°

d. 146°



2 A geometric figure scaled by factor $k = 2$ has an original side length of 8 cm. What is the corresponding image length?

a. 14 cm

b. 16 cm

c. 24 cm

d. 32 cm

3 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 4)$. What is its y-coordinate when $x = 1$?

a. 4

b. 9

c. 7

d. 10

Procedural Fluency & Algebraic Manipulation

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4 In $\triangle ABC$ studying coordinate methods and triangle congruence, $m\angle A = 34^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 94°

b. 104°

c. 86°

d. 146°

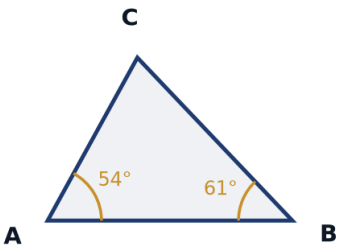
5 A geometric figure scaled by factor $k = 3$ has an original side length of 8 cm. What is the corresponding image length?

a. 21 cm

b. 24 cm

c. 32 cm

d. 72 cm



6 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 4)$. What is its y-coordinate when $x = 2$?

a. 7

b. 12

c. 10

d. 13

Graphical & Geometric Interpretation

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7 In $\triangle ABC$ studying coordinate methods and triangle congruence, $m\angle A = 34^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 87°

b. 97°

c. 79°

d. 146°

8 A geometric figure scaled by factor $k = 4$ has an original side length of 8 cm. What is the corresponding image length?

a. 28 cm

b. 32 cm

c. 40 cm

d. 128 cm

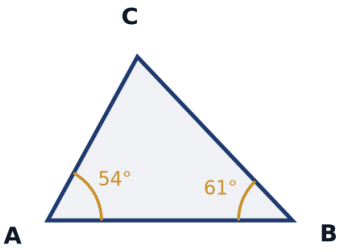
9 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 4)$. What is its y-coordinate when $x = 3$?

a. 10

b. 15

c. 13

d. 16



Exam-Style Problem Solving & Modeling

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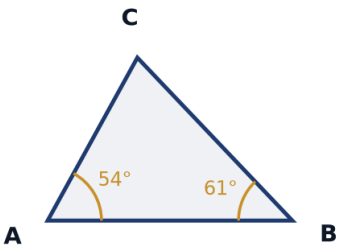
10 In $\triangle ABC$ studying coordinate methods and triangle congruence, $m\angle A = 34^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 80°

b. 90°

c. 72°

d. 146°



11 A geometric figure scaled by factor $k = 5$ has an original side length of 8 cm. What is the corresponding image length?

a. 35 cm

b. 40 cm

c. 48 cm

d. 200 cm

12 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 4)$. What is its y-coordinate when $x = 4$?

a. 13

b. 18

c. 16

d. 19

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